

Lattice Distortions and Ion Conduction Mechanisms in NASICON Solid Electrolytes

1. Scientific background

Sodium solid-state electrolytes (SSEs) are key to developing safer, high-energy-density sodium batteries [1]. These include Na superionic conductor (NASICON)-type materials, which combine high ionic conductivity with favourable electrochemical stability [1, 2], with general formula $\text{Na}_x\text{M}_2(\text{AO}_4)_3$, where M is a transition or main group metal and AO_4 is a polyanion [3]. The canonical formulation is $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ (NZSP), which has been shown to reach ionic conductivity of up to 5 mS/cm [4]. However, ionic conductivity must improve by at least an order of magnitude to rival liquid electrolytes [5]. It is now widely held that achieving this requires understanding how chemical modifications alter atomic-scale dynamics [5, 6].

NASICONs also exhibit interesting fundamental science phenomena, making them a suitable platform for studying ion-transport physics under different chemistries [1]. For example, the Na sites (Na1 & Na2 in Fig. A(i)) are linked by geometric ‘bottlenecks’, forming a 3D network [5]. The bottleneck volume, Na content, metal (M) and polyanion (AO_4) types, and mixing ratios define the Na site energy distribution. It is believed this energy distribution defines the Na-ion conduction mechanism, where NASICON with a larger Na site energy difference would prefer single-ion hopping (SIH), in contrast to occupancy-conserved hopping (OCH). This is shown by the activation energies in Figures A(ii) and A(iii), respectively [5]. Furthermore, lattice distortions due to multiple M species mixing improves ionic conductivity by widening the Na sites energy overlap, as shown in Figure B(i); the effect these metal species have on ionic conductivity in real NASICON-type materials is shown in Fig. B(ii) [7].

Better design rules for faster solid-state electrolytes will be established by improving understanding of the dynamics of SIH and OCH, and by quantifying the distortion during conduction in a controlled variation of NASICON compositions [5]. However, such variation in conduction mechanisms and their physics has not been probed directly. We therefore propose to directly measure fluctuations in NASICON samples on the atomic spatial scale and 10^{-2} – 10^3 s time scale by performing x-ray photon correlation spectroscopy (XPCS) on the intense (116), (104) and (113) Bragg peaks on a range of NASICON samples.

2. Motivation for the proposal, expected results and impact

Understanding how Na-ion conduction shifts between different mechanisms, such as SIH and OCH [5], is key to optimizing NASICON materials. By clarifying how composition influences the dominant pathway, we can maximise ionic conductivity while maintaining other critical properties like electrochemical stability. Establishing methods to distinguish between these mechanisms is thus key for defining the limits of design strategies, such as those that rely on OCH pathways [7]. XPCS can directly probe local lattice distortions during ion transport [8], an effect predicted by density functional theory to strongly influence conductivity [7]. Our previous beamtime at P10 demonstrated the efficacy of this technique on a different Na SEE compound, the glassy Na_3PS_4 , by performing XPCS scans over a range of temperatures. The soon-to-be published results in Fig. C(i) & C(ii) respectively show temperature-dependent $g^{(2)}(\mathbf{Q}, \tau)$ and fitting of collective fast timescale τ and β parameters, and the two-time correlation plot in Fig. C(iii) reveals non-linear dynamics present within the system.

For this experiment we intend to probe dynamics in a similar way on our NASICON samples. By varying the metal species M, which will change the 6b and 18e Na-site energy difference [5], while also keeping the temperature fixed at RT, we expect to see a shift in the $g^{(2)}(\mathbf{Q}, \tau)$ decay time. In addition, we expect decay exponent β to depend on the Na-sites energy difference, with NASICONs that have a larger 6b and 18e Na-site energy difference (SIH mechanism) having a higher β , due to a more homogeneous lattice change during relaxation, and those with smaller energy difference (OCH) having a smaller β [5]. The precise relationship between ion mobility and correlation and ionic conductivity in solid-state electrolytes remains an open and unanswered question in the development of solid-state batteries. XPCS is a valuable tool for directly observing these fluctuations at the nanoscale and is now starting to be employed as part of the toolkit for characterisation of these compounds, alongside more established neutron-based techniques [9, 10]. By demonstrating the efficacy of XPCS in this context and laying out an analysis pipeline, the published results of this work are expected to have a significant impact on the study of SSEs and other ionic materials with XPCS.

3. Own previous work

Our team is a collaboration of researchers from two groups who are part of the US Department of Energy’s research hub, the [Energy Storage Research Alliance](#), comprised of 3 US national labs and 11 universities, with the aim of

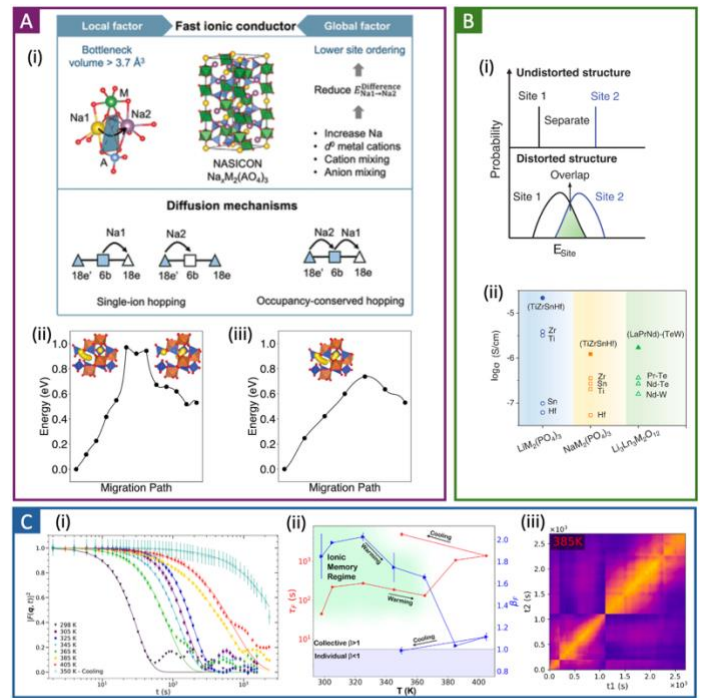


Fig. A: The structure of NASICON and the factors that make it a fast ionic conductor (i); and calculated energy landscape of NASICON $\text{Na}_3\text{Sb}_2(\text{SiO}_4)_3$ following the SIH (ii) and OCH (iii) mechanisms [5]. B: Schematic showing how local distortions create overlapping site-energy distributions (i); and ionic conductivities of NASICON-type materials and their single-metal counterparts at room temperature [7]. C: Unpublished wide-angle XPCS results on the SSE Na_3PS_4 from PETRA-P10 showing g_2 autocorrelation functions (i), T-dependent fast timescale fitting parameters, and a two-time autocorrelation function (iii).

developing the next generation of energy storage. The Shpyrko group at UC San Diego has performed coherent x-ray experiments at numerous synchrotron and XFEL facilities [8], including as-yet unpublished recent wide-angle Bragg XPCS studies at APS-U and PETRA III. Group members have performed work at P10, including successful proposals I-20211701, I-20221009 and I-20230664. The Ceder group (represented here by PhD candidate Bernadus Rendy) at UC Berkeley has established itself as a pioneer in battery research, especially in understanding NASICON synthetic accessibility, ionic conduction mechanisms, and their design rules through both computational and experimental approaches [3, 7, 11].

4. Experimental plan

a) Justification of beamline requested

From our experience at beamtimes around the world, beamline P10 has the most developed and user-friendly XPCS setup. It also has the necessary focusing and coherent flux required to measure individual micron-sized particles of the NASICON samples, and detector geometry to capture Bragg peaks at large q-values.

b) Justification of requested number of shifts

Initial setup should take no longer than one shift. Considering overhead and sample changes, we then expect to record around one XPCS scan per hour for the remaining time. With 10 NASICON samples, 3 peaks per sample and up to 3 readings per peak, we are requesting 18 shifts.

c) Data handling and computing

Image sequences will be captured by the Eiger 4M camera at a maximum frame rate of 0.75 kHz. Preliminary data analysis will be performed on-the-fly using the MATLAB-based software package XPCSGUI. This allows us to assess whether the measurement has been successful and can change Bragg peak position or move to the next sample. Typical acquisitions are around 1/2 GB, and we expect to acquire around 100 scans, meaning around 50 GB of data storage is needed.

d) Samples & safety related issues

The samples will be the oxide-based electrolyte NASICON, suspended in the 2-part epoxy resin Stycast, formed into an approximately 3-5 mm diameter, 2 mm thick disc. There are no safety hazards associated with handling the samples or placing them inside a vacuum and temperature-controlled environment.

e) technical information (if applicable)

Each sample will be mounted in a sample stage using the standard XPCS configuration on the 4-circle SAXS/WAXS setup at P10. No temperature control or external stimuli like a laser or power supply will be required. The only foreseen difficulty expected is locating Bragg peaks from the suspended particles. However, we know the beamline apparatus is well equipped for this task.

5. References related to the proposal scientific text

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